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How customers cope with service failure? A study of brand reputation and customer satisfaction

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ABSTRACT

In this paper we examine the different coping mechanisms used by customers when they confront service failure. It was found that the coping mechanism used by customers changes depending on the severity of the service failure. Further, it was observed that brand reputation moderates the relationship between severity of service failure and coping strategies, customer satisfaction and behavioral intentions under different conditions. We conducted an experimental study in an airline travel context. The data supports the proposed model.

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1. Introduction

Failures in service delivery are inevitable and recovery of such encounters represents a significant challenge for service firms. When service failure occurs, customers experience disconfirmation and feel displeased (Smith, Bolton, & Wagner, 1999). Company's reputation in conjunction with its service offering determines customer's expectation. Prior studies report that customers maintain different levels of expectations for high and low reputed brands (Roggeveen, Bharadwaj, & Hoyer, 2007). For high reputed brands, customers have high expectations, and view them as delivering superior services. Research findings have confirmed that customers positively relate to brand reputation and this affects their perception of service performance and benefits (Brodie, Whittome, & Brush, 2009). In such instances, how brand reputation affects the evaluation of service failure is an important managerial issue. To date, little research has focused on the role of brand reputation in assessing the service failure outcomes.

Service failure elicits an internalization process of coping by which the customers attempt to mitigate the stress associated with negative experiences. During service failure, customers engage in a profile of coping strategies such as active coping, avoidance, and support seeking to overcome the undesirable state (Duhachek & Kelting, 2009; Lazarus & Folkman, 1984). However, the stress experienced is contingent upon the expectations customers have of the service provider (Buchanan, Simmons, & Bickart, 1999). This suggests that customers might use different coping strategies depending upon the expectations associated with the service brand that has failed to deliver. However, there is little systematic research on the linkages between brand reputation, customer coping strategies, and service failure outcomes. To fill this research gap, the current study extends the service failure literature to examine the different coping strategies that customers use in managing service failure, and to determine the role of brand reputation in coping with the service failure.

In this paper we attempt to address a number of issues pertaining to our understanding of services and service failures. First, we examine the role of coping in service failure encounters. Although prior literature proposes various coping strategies to deal with stress from economic crisis, natural disasters, health-related outcomes and depression (Scheier & Carver, 1985; Skoufias, 2003), it is unclear whether they are applicable in a service failure context. Further, as consumers use distinct appraisal processes in evaluating stressful encounters, we expect evaluation of service failures to be associated with different coping strategies. Second, brand reputation impacts the customers' expectations of service performance. In this study, we show that brand reputation and the expectations thereof, influence the evaluation of service failure and customer coping strategies. Finally, this study examines the mediating role of customer coping strategies in the relationship between service failure, brand reputation, and behavioral outcomes. While prior studies have empirically shown the buffering effects of brand reputation (Brady, Cronin, Fox, & Roehm, 2008; Huang,

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2011), the process by which this occurs is not yet fully understood. Few studies that have examined the role of brand reputation using attribution perspective have produced mixed findings (Hess, 2008; Liao & Cheng, 2013). Consequently, we use coping perspective to understand the process by which brand reputation affects the service failure evaluation and its outcomes. This has important implications for the firm in understanding the customers' evaluation process and design appropriate recovery strategies.

2. Theoretical framework and hypotheses development

This research investigates the role of brand reputation in coping with service failure and the mediating role of coping strategies in influencing satisfaction and behavioral intentions. While the focus on research until recently has been on service failure (Mattila, 2004; Smith et al., 1999), the different coping strategies customers use to deal with the service failure have not yet been thoroughly examined. Further, Erdem, Swait, and Louviere (2002) contend that brands serve as an effective signal to communicate information and quality expectations. Thus, we study the coping strategies customers use to manage service failures occurring in the case of high reputed brands and low reputed brands. Fig. 1 presents the conceptual framework of the study.

2.1. Consumer coping strategies

Coping refers to the actions or thoughts that people use in dealing with stressful encounters. As Folkman, Lazarus, Gruen, and DeLongis (1986) point out, coping is the "constantly changing cognitive and behavioral efforts to manage specific external and/or internal demands that are appraised as taxing or exceeding the resources of the person (p. 993)." The coping strategies are aimed at changing the distressed customer-environment relationship by either altering the relationship and/or by regulating the emotions (Lazarus & Folkman, 1987). Most coping research has been based on Folkman and Lazarus's (1985) seminal work that classified coping into two distinct but complementary strategies: emotion-focused coping and problem-focused coping. A similar framework of coping was proposed by Duhachek (2005) consisting of expressive coping, active coping and avoidance/denial coping. While these studies have conceptualized coping with greater specificity, other researchers have argued for diverse coping strategies that are more parsimonious, empirically derived and theoretically rich (Carver, Scheier, & Weintraub, 1989; Yi & Baumgartner, 2004). According to Lazarus (2006), the distinction of coping as problem-focused and emotion-focused is too simple, and inaccurate of the way customers use coping strategies to manage stressful encounters. Thus, in this study we propose a multidimensional measure consisting of eight coping strategies.

2.1.1. Action coping (AC)

Consumers who cope with service failures through action coping think about ways to make things better, devise a plan of action, and try to resolve the problem (Duhachek, 2005). They are likely to ruminate about the stressful encounter and possibly attribute the failed encounter to the service provider (Strizhakova, Tsarenko, & Ruth, 2012). This directs the customers to directly confront the service provider to express their displeasure and seek redress.

2.1.2. Rational thinking (RT)

Consumers, who engage in rational thinking, view the service failure from a realistic perspective and focus on the causes by systematically examining the problem (Lazarus & Folkman, 1987). They might attribute the failure as controllable and try to engage in problem solving (Jin & Hong, 2010). Consumers using rational thinking exhibit restraint from acting rashly and avoid showing others how they feel about the negative experience.

2.1.3. Positive thinking (PT)

Positive thinking is a constructive coping response where consumers see an opportunity to learn from the stressful encounters (McCrae, 1984). They perceive the reasons for service failure as uncontrollable and unpredictable and readily forgive the service provider by continuing their patronage (Johnson & Rapp, 2010). The customers accept that the problem could have been worse and concentrate on the good to facilitate positive appraisals of the encounter.

2.1.4. Emotional venting (EV)

Consumers who experience a stressful encounter may try to vent their feelings to get the attention of the service provider. It is a confrontative response in which customers ruminate about the failure and regulate their emotions by expressing their anger or displeasure to the service provider. Parlamis (2012) showed that customers are more likely to vent their anger when they emphasize internal and controllable attribution for the failed encounter.

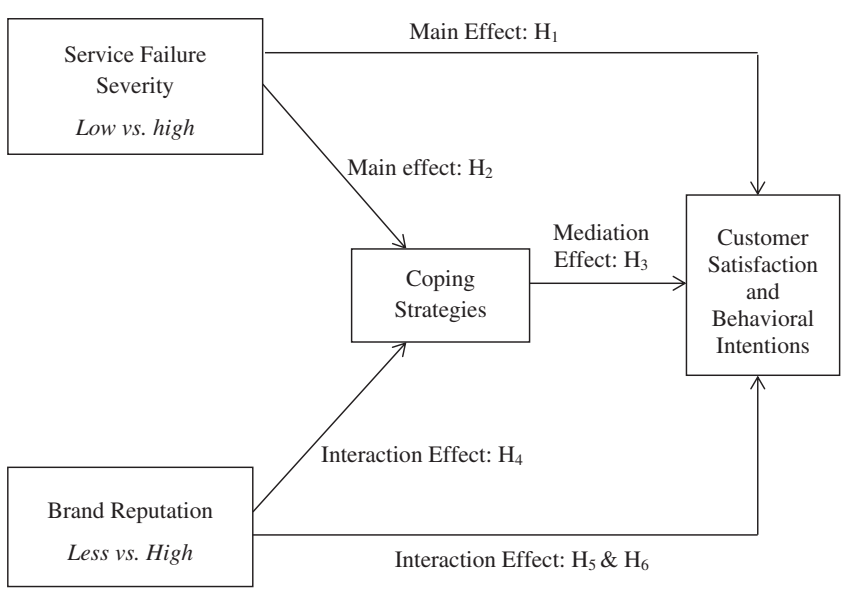

Fig. 1. Conceptual framework of the effects of magnitude of failure and reputation on coping strategies and satisfaction.

2.1.5. Instrumental-support seeking (IS)

Consumers who cope with instrumental-support seeking try to seek advice from friends or acquaintances with similar experiences (Yi & Baumgartner, 2004). Consumers adopting instrumental-support coping perceive the service failure as controllable (Georgiou & Stavrinides, 2008) and seek assistance or advice from other customers or from one's social network regarding the complaint process and/or the redress.

2.1.6. Emotional-support seeking (ES)

Consumers aim at managing the service failure encounters by expressing their feelings to others to seek comfort. It involves obtaining moral support or sympathy from others. It differs from instrumental-support seeking as customers rely on others as outlets for the expression of their feelings (Yi & Baumgartner, 2004). Customers attribute the service failure to the situational factors (Gelbrich, 2010) and try to regulate their emotional distress by disclosing their feelings to others.

2.1.7. Avoidance (AV)

Avoidance is a passive way of coping in which customers distance themselves from the problem to restore their emotional balance (Strizhakova et al., 2012). Thus, consumers cope with service failure encounters by indulging in other things to distract them from the problem. They attribute the service failure to uncontrollable and stable reasons (Yi & Baumgartner, 2004) and thus do not patronize the service provider anymore in the future.

2.1.8. Denial (DEN)

Consumers may cope by pretending that the service failure encounter never happened. Denial serves the function of managing one's emotions (Duhachek, 2005). Customers exhibit greater rumination and unresolved emotional distress with denial coping strategy (Strizhakova et al., 2012).

2.2. The impact of failure severity on consumer coping strategies

Failure severity refers to the magnitude or intensity of the service failure. Customers who experience severe failure are likely to perceive greater loss, evaluate the service unfavorably and report dissatisfaction. They show aversion towards continuing relationship and indulge in negative word-of-mouth towards the service provider (del Río-Lanza, Vázquez-Casielles, & Díaz-Martín, 2009; Kalamas, Laroche, & Makdessian, 2008; Smith et al., 1999).

When customers perceive the service failure as severe, they may deliberately take efforts to reduce the greater perceived loss and negative emotions. They ponder on the causes and consequences of the failure and actively engage in effortful actions to resolve the situation. Further, severe failures evoke anger which intensifies expressive tendencies and retaliatory behaviors (Bonifield & Cole, 2007). Thus, we expect customers to use action coping, emotional venting, and instrumental-support seeking to manage high severe service failures. On the contrary, when customers experience less severe failures, they evaluate the potential loss as low and might ignore the negative emotions. In such situations customers are less likely to blame the service provider and try to remove themselves from the stressful situation (Gelbrich, 2010). This would suggest the use of coping strategies such as rational thinking, positive thinking, emotional-support seeking, avoidance, and denial. Thus, we hypothesize that:

H1. Service failure has a significant main effect on customer satisfaction and behavioral intentions such that service failure severity is negatively related to (a) customer satisfaction, and (b) behavioral intentions.

H2. Service failure has a significant main effect on customer coping strategies such that service failure severity is positively related to a) action coping, b) instrumental support seeking, c) emotional venting, and negatively related to d) rational thinking, e) positive thinking, f) emotional support seeking, g) avoidance and h) denial.

2.3. The mediation effect of coping strategies

In the present study, we propose that appraisal of the service failure triggers different coping strategies that influence the evaluation of the service encounter. Gabbott, Tsarenko, and Mok (2011) showed that when customers employ coping strategies such as action coping or instrumental support seeking, they are likely to evaluate the service encounter as unfavorable. These coping strategies are associated with greater distress and anxiety as customers pay greater attention to the causes of failure and the possible solutions (Folkman et al., 1986). Thus, it is expected to result in lower customer satisfaction and behavioral intentions. On the other hand, the use of coping strategies such as positive thinking, rational thinking, and emotional-support seeking results in reinterpretation of the stressful encounter (Gabbott et al., 2011). This results in a more favorable evaluation of service encounter. Thus, it is expected to result in greater customer satisfaction and behavioral intentions. Similarly, when customers use avoidance and denial they ignore the negative experiences. However, when customers inadequately address the lingering negative emotions it could lead to lower customer satisfaction (Strizhakova et al., 2012). Thus, we hypothesize that:

H3. The coping strategies mediate the relationship between failure severity and customer satisfaction.

2.4. The moderating role of brand reputation on coping strategies and customer outcomes

Brand reputation refers to the customers' perception of service quality associated with the brand name (Selnes, 1998). Prior studies indicate that favorable brand reputation protects the firm during the crisis by creating a buffer or shield (Hess, 2008). Customers attribute the firm to be less responsible for the crisis and give the benefit of doubt to the organization. They discount the negative information and focus on the positive aspects associated with the firm. In a study, Huang (2011) successfully showed that brand equity moderates the relationship between service recover and satisfaction, such that for high equity brands, distributive justice results in higher recovery satisfaction than low equity brands. It was argued that the strong and favorable quality associations with high equity brands help customers attribute the service failure as temporary and unstable, resulting in lower dissatisfaction. Contrarily, Hess (2008) argued that excellent reputation does not buffer the negative effects of service failure as it did not reduce customers' attribution of failure. Similarly, Brady et al. (2008) suggested that customer satisfaction and behavioral intentions are dependent on the evaluation of the recovery efforts, irrespective of the equity attached with the brand. These mixed findings suggest the need for further work to evaluate the role of brand reputation in customer reactions to service failure.

The present study uses key insights from expectancy-disconfirmation theory and uncertainty management theory in hypothesizing the role of brand reputation on service failure outcomes. The expectancy-disconfirmation theory suggests that customers form expectations of service performance prior to purchase (Oliver & Burke, 1999). When customers encounter service failure, negative disconfirmation occurs as product performance fails to meet the expectations. For high reputed brands, customers have high expectations that result in greater negative disconfirmation and dissatisfaction with the service failure. They might experience 'shock' or 'betrayal' on encountering service failure in high reputed brands (Brady et al., 2008). This betrayal or shock displaces the positive attitudes and beliefs associated with the brands resulting in unfavorable evaluation of service failure (Mattila, 2004).

Similarly, uncertainty management theory posits that people tend to react more unfavorably when they face uncertainty in the environment. They feel anxious and distressed and try to find comfort in related or unrelated experiences (Lind & Van den Bos, 2002). Furthermore, they indulge in disparaging and antisocial behaviors towards the organization. Consequently, when service failures occur in highly reputed brands, customers experience greater uncertainty motivating them to cope by seeking salient information to reduce the anxiety and distress (Kashyap, 2001).

Along somewhat similar lines, Anderson (1998) reported negativity bias in relation to usefulness and salience of negative information. When customers are confronted with information inconsistency, the negative information is perceived as being more diagnostic and relevant in invoking biased processing and generating counterarguments (Ahluwalia, Unnava, & Burnkrant, 2001). This biased processing can be explained by "the novelty of the negative information, its lower ambiguity, or the greater discrepancy with the general positivity of the typical" (Wojciszke, Brycz, & Borkenau, 1993, p. 327). Thus, when customers experience high severe service failures in highly reputed brands, they evaluate the negative experience as novel outcome resulting in greater disconfirmation and lower satisfaction. For less severe failures, we propose that customers might ignore the negative information as they expect occasional mild failures to occur (Hess, 2008). This results in greater satisfaction for less severe failures in high reputed brands than less reputed brands.

There is ample support in the service literature on how brand reputation positively affects behavioral intentions (Gounaris & Stathakopoulos, 2004). The positive quality associations of highly reputed brands provide customer cues that the service failure is an aberration. They perceive that the failure would not happen in the future and thus might exhibit higher behavioral intentions (Brady et al., 2008). Thus, we hypothesize that:

H4. Brand reputation moderates the relationship between failure severity and coping strategies.

H5. Brand reputation moderates the relationship between failure severity and customer satisfaction such as for less severe failures customers experience greater satisfaction for high reputed brands than less reputed brands.

H6. Brand reputation moderates the relationship between failure severity and behavioral intentions such as for high reputed brands customers experience greater behavioral intentions than less reputed brands for both high and less severe failures.

3. Methodology

3.1. Research design and sample

This study adopted a 2 (failure severity: high vs. less) × 2 (brand reputation: high vs. less) scenario-based experimental design. Airline travel was chosen as the study context as it was widely used in prior studies to examine customer response to service failures (Karande, Magnini, & Tam, 2007). A total of 261 graduate students participated in the study in exchange for extra course credits. The average age of the respondents was 22.7 years, with 57.5 percent of the participants' male and 42.5 percent female. Almost all participants (98%) had traveled by air in the last 6 months.

3.2. Procedure

The scenarios developed, based on the online customer reviews, described the reputation of the airline service provider following which the service failure was presented. Appendix 1 presents the scenarios used in the study. After reading the scenario each participant filled a paper-and-pencil survey. Following the experiment session, the participants were debriefed about the research purpose and thanked.

3.3. Manipulation checks

The manipulation of service failure condition was checked by asking the participants to respond to the three statements: the problem encountered in the scenario is a "Major issue," "High severe" and "Major inconvenience." The manipulation of brand reputation condition was assessed with the three statements: the reputation of the airline in the scenario "is a popular brand," "has high esteem" and "has status." Respondents rated these questions on a 10-point Likert scale ranging from "1" strongly disagree to "10" strongly agree.

3.4. Measures

The eight coping strategies were measured using 36 items adopted from Duhachek's (2005) study. Customer satisfaction and behavioral intentions were measured using 3 items adopted from Gabbott et al.'s (2011) study. Satisfaction was measured by a single item "overall satisfaction with the airline services" and two items "how likely would you use this airline again" and "how likely are you to recommend this airline to your friends?" were used to measure behavioral intentions. All items were measured on a seven-point Likert scale ranging from "1" strongly disagree to "7" strongly agree.

4. Results

4.1. Manipulation and realism checks

To assess the manipulation of stimuli, ANOVA was performed with failure severity and brand reputation as fixed factors on the means of the three service failure questions (Cronbach's $\alpha = 0.82$) and three brand reputation questions (Cronbach's $\alpha = 0.95$) separately. Results showed a significant main effect of service failure condition on failure severity evaluation ($F_{(1, 257)} = 153.1$, $p < 0.01$). The participants in less severe condition ($M = 5.74$) indicated a significantly lower level of failure severity than those in the high severe condition ($M = 8.88$) ($F_{(1, 257)} = 153.1$, $p < 0.01$). Similarly, the participants in high reputed brand condition ($M = 7.60$) perceived a significantly greater level of brand reputation than those in less reputed brand condition ($M = 3.34$) ($F_{(1, 257)} = 442.30$, $p < 0.01$). These results indicated that the manipulations were perceived as intended. The perceived realism of scenario was high ($M = 6.12$, S.D. $= 2.20$) suggesting that participants found the reputation and failure scenarios believable.

4.2. Construct validity

Confirmatory factor analysis (CFA) was carried out to assess the validity of the eight coping strategies. All 36-items for the eight factor coping construct were included in the measurement model, but the initial CFA result was not accepted because of poor model fit statistics: $\chi^2 = 973.56$, $df = 493$, goodness of fit index (GFI) $= 0.82$, comparative fit indices (CFI) $= 0.87$, and root mean square error of approximation (RMSEA) $= 0.062$. After dropping 10 items from the measurement model, the CFA results showed a good model fit with $\chi^2 = 487.51$, $df = 274$, GFI $= 0.90$, CFI $= 0.93$ and RMSEA $= 0.055$. These results provide support for the eight-factor coping model. The convergent validity was supported by the significant item-factor loadings ($p < 0.01$) which ranged from 0.72 to 0.90 (see Table 1). Moreover, the average variance extracted (AVE) for each factor was well above the recommended threshold levels of 0.70 (Fornell & Larcker, 1981). The construct reliability (CR) ranged from 0.83 for emotional venting to 0.92 for action coping, all well-above the recommended threshold level of 0.70 (Hair, Black, Babin, Anderson, & Tatham, 2006).

Table 1
Confirmatory factor analysis on coping scale.

	Λ	AVE	CR
Action coping (AC)			
AC1: I will concentrate on ways to solve the problem	0.80	0.68	0.92
AC2: Try to make a plan of action	0.83		
AC3: Generate potential solutions to solve the problem	0.83		
AC4: Think about the best way to handle the problem	0.88		
AC5: Concentrate my efforts on doing something about the problem	0.79		
Rational thinking (RT)			
RT2: Try to step back from the situation and be objective	0.70	0.56	0.84
RT3: Try to control my emotions	0.79		
RT4: Try to restrain my feelings from controlling my actions	0.79		
RT5: Use restraint to avoid acting rashly with the airline	0.72		
Positive thinking (PT)			
PT1: Try to look at the brighter side of the problem	0.86	0.76	0.91
PT2: Focus on positive aspects of the problem	0.90		
PT3: Look for the good in what happened	0.86		
Emotional venting (EV)			
EV3: Delve into my feelings to understand of them	0.82	0.63	0.83
EV4: Realize that my feelings are valid and justified	0.83		
EV6: Acknowledge my emotions	0.72		
Instrumental support (IS)			
IS1: Ask friends with similar experience what they did	0.79	0.66	0.85
IS2: Try to get advice from someone about what to do in such cases.	0.86		
IS3: Have a friend assist me in solving the problem so that I can reach my destination.	0.78		
Emotional support (ES)			
ES1: Tell others how I feel	0.81	0.62	0.87
ES2: Rely on others to make me feel better	0.77		
Avoidance (AV)			
AV1: Try to take my mind off of this experience with the airline by doing other things	0.75	0.67	0.86
AV2: Distract myself to avoid thinking about this experience	0.86		
AV3: Avoid thinking about this experience	0.85		
Denial (DEN)			
DEN1: Deny that this encounter ever took place	0.88	0.77	0.91
DEN2: Refuse to believe that the problem had occurred	0.87		
DEN3: Pretend that this never happened	0.89		

Λ — Standardized factor loadings; AVE — Average variance extracted and CR — Construct reliability.

Discriminant validity was examined with a variance-extracted test, where the average variance extracted estimate of a construct was assessed with the square of the correlation between the constructs (Fornell & Larcker, 1981). As shown in Table 2 discriminant validity was demonstrated as the variance-extracted estimates are greater than the corresponding squared correlations.

To further assess the validity of the eight-factor model, the fit indices of competing models; one factor model, two factor model proposed by Lazarus and Folkman (1984) and Holahan and Moos (1987), and three-factor model proposed by Duhachek (2005) were examined. As shown in Table 3, the eight-factor model provided the best fit of the data. As a result, eight coping factors are considered for the present study.

4.3. Hypotheses testing

4.3.1. Testing H1, H5 and H6

Multivariate analysis of covariance (MANCOVA) followed by univariate comparisons was used to test the hypotheses proposed. The dependent variables were customer satisfaction and behavioral intentions. Failure severity and brand reputation were used as fixed factors. Gender and age, initially included as covariates, were dropped from further analysis as they did not significantly affect the relationship (gender: $F = 0.31$, $p = 0.16$; age: $F = 0.74$, $p = 0.72$).

Results of multivariate analysis of variance as presented in Table 4 showed significant main effects of failure severity (Wilk's $\Lambda = 0.87$, $F = 19.25$, $p < 0.01$, $\eta_p^2 = 0.13$), brand reputation (Wilk's $\Lambda = 0.89$, $F = 15.69$, $p < 0.05$, $\eta_p^2 = 0.11$), and a significant two-way interaction between failure severity and brand reputation (Wilk's $\Lambda = 0.97$, $F = 3.46$, $p < 0.05$, $\eta_p^2 = 0.03$). Examination of the cell means indicates that respondents in high severe group reported lower levels of satisfaction ($M = 2.20$) and lower levels of behavioral intentions ($M = 2.16$) than did the respondents in the less severe group (satisfaction; $M = 2.78$, behavioral intentions; $M = 3.18$). This supports Hypotheses H1(a) and H1(b).

The univariate results indicate significant interaction effect of failure severity and brand reputation on satisfaction ($F = 5.84$, $p < 0.05$). The simple slope analysis for less severe failure group showed that high reputation group ($M = 3.18$) reported higher levels of customer satisfaction than less reputation group ($M = 2.31$) (simple slope $= 0.26$, $p < 0.01$). Cell means of high severe failure did not reveal difference in customer satisfaction for the less reputation group ($M = 2.26$) and high reputation group ($M = 2.15$), (simple slope $= -0.03$, $p = 0.72$). This provides support for H5. The interaction effect is presented in Fig. 2.

For behavioral intentions, the interaction effect of failure severity and brand reputation was marginal ($F = 2.80$, $p = 0.09$). Simple slope analysis revealed significant difference between less severe-high reputed group ($M = 2.55$) and less severe-less reputed group ($M = 3.71$), simple slope $= 0.41$, $p < 0.01$. Similarly for high failure severity group, respondents evaluated greater behavioral intentions in high reputed group ($M = 2.47$) than less reputed group ($M = 1.85$), simple slope $= 0.25$, $p < 0.01$. This provides marginal support for H6. Fig. 3 plots the interaction effect on behavioral intentions.

4.3.2. Testing H2 and H4

Results of MANCOVA showed significant main effect of failure severity (Wilk's $\Lambda = 0.74$, $F = 10.81$, $p < 0.01$, $\eta_p^2 = 0.26$) and interaction effect of failure severity and brand reputation (Wilk's $\Lambda = 0.92$, $F = 2.85$, $p < 0.01$, $\eta_p^2 = 0.08$) on coping strategies. The univariate results presented in Table 5 show significant main effect of failure severity on

Table 2
Descriptive statistics, correlation and discriminant validity of coping scale.

	Mean	SD	AC	RT	PT	EV	IS	ES	AV	DEN
Action coping	4.92	1.33	(0.68)	0.00	0.00	0.00	0.09	0.00	0.00	0.00
Rational thinking	4.10	1.21	0.05	(0.56)	0.30	0.21	0.07	0.02	0.15	0.06
Positive thinking	3.57	1.58	−0.07	0.55*	(0.76)	0.29	0.04	0.06	0.11	0.17
Emotional venting	3.52	1.31	0.07	0.46*	0.54*	(0.63)	0.05	0.14	0.11	0.14
Instrumental support	4.23	1.48	0.30*	0.27*	0.21*	0.23*	(0.66)	0.14	0.03	0.00
Emotional support	3.61	1.61	0.03	0.16**	0.25*	0.38*	0.38*	(0.62)	0.08	0.18
Avoidance	4.21	1.41	−0.03	0.39*	0.34*	0.34*	0.17*	0.29*	(0.67)	0.09
Denial	2.66	1.54	−0.07	0.25*	0.42*	0.37*	0.06	0.43*	0.30*	(0.77)

The diagonal values represent average variance extracted estimates of the constructs. The values in the lower diagonal of the table present the correlations between the constructs while the values in the upper diagonal of the table present squared correlations between the constructs.
* $p < 0.01$.
** $p < 0.05$.

Table 3
Confirmatory factor analysis model fit comparisons.

Model	Chi-square	*df*	Chi-square difference
Null	3018.27	325	
1-Factor model	1956.18[a]	299	1062.1*
8-Factor correlated	487.51[b]	274	1468.67*
2-Factor correlated model (Lazarus & Folkman, 1984)	602.11[c]	289	114.6*
3-Factor correlated model (Duhachek, 2005)	624.28[d]	288	136.77*
2-Factor correlated model Holahan and Moos (1987)	628.61[e]	290	141.1*

	GFI	CFI	TLI	RMSEA
1-Factor model	0.56	0.38	0.33	0.147
8-Factor correlated model	0.88	0.93	0.93	0.055
2-Factor correlated model (Lazarus & Folkman, 1984)	0.85	0.88	0.87	0.065
3-Factor correlated model (Duhachek, 2005)	0.84	0.87	0.86	0.068
2-Factor correlated model Holahan and Moos (1987)	0.84	0.87	0.86	0.068

Note: *df*is degrees of freedom, GFI is Goodness of Fit, CFI is Comparative Fit Index, and TLI is Tucker Lewis Index.
* $p < 0.01$.
[a] Chi-sqaure differences represent comparisons of 1-factor model with null-model.
[b] Chi-square differences represent comparisons of 8-factor model with1-factor model.
[c] Chi-square differences represent comparisons of 2- factor model with 8-factor model.
[d] Chi-square differences represent comparisons of 3-factor model with 8-factor model.
[e] Chi-square differences represent comparisons of 2-factor model with 8-factor model.

coping strategies of action coping ($F = 54.77$, $p < 0.01$, $\eta_p^2 = 0.18$). The high severe group (M = 5.43) reported greater levels of action coping than did the less severe group (M = 4.34), supports hypothesis H2(a). H2(b) was supported as the high severe group (M = 4.42) reported greater levels of instrumental-support seeking than did the less severe group (M = 4.02), $F = 4.67$, $p < 0.05$, $\eta_p^2 = 0.02$. We could not find support for H2(c) as no difference was observed in using emotional venting across high severe (M = 3.69) and less severe groups (M = 3.34), $F = 1.00$, $p = 0.32$, $\eta_p^2 = 0.00$. Supporting H2(d), we found that less severe group (M = 4.30) reported higher levels of rational thinking coping strategy than did the high severe group (M = 3.93), $F = 5.44$, $p < 0.05$, $\eta_p^2 = 0.02$.

Examination of univariate results show that respondents in the less severe group (M = 3.97) reported significantly higher levels of positive thinking than did the high severe group (M = 3.21), $F = 15.08$, $p < 0.01$, $\eta_p^2 = 0.06$. This negative relationship provides support for H2(e). We did not find support for H2(f) as high severe group (M = 3.61) and less severe group (M = 3.62) did not differ in use of emotional support seeking, $F = 0.02$, $p = 0.89$, $\eta_p^2 = 0.00$. For avoidance, the respondents in less severe group (M = 4.58) reported higher disengagement with the service failure than did the high severe group (M = 3.89), $F = 16.33$, $p < 0.01$, $\eta_p^2 = 0.06$, providing support for H2(g). Finally, the less severe group (M = 2.90) reported higher levels of denial than did the high severe group (M = 2.44), $F = 5.46$, $p < 0.05$, $\eta_p^2 = 0.02$. This provides support for H2(h).

The univariate results indicated significant interaction effect of failure severity and brand reputation on action coping (F = 7.44, $p < 0.01$, $\eta_p^2 = 0.03$), rational thinking (F = 4.72, $p < 0.05$, $\eta_p^2 = 0.02$), positive thinking (F = 5.78, $p < 0.05$, $\eta_p^2 = 0.02$) and emotional venting (F = 5.76, $p < 0.05$, $\eta_p^2 = 0.02$). Examination of cell means indicates that in less severe condition, respondents reported greater action coping for high reputed brand (M = 4.60) than for less reputed brand (M = 4.03) (simple slope = 0.21, $p < 0.05$). However, no significant difference was observed in action coping for high severe failures between less reputed (M = 5.55) and high reputed brands (M = 5.30) (simple slope = −0.11, $p = 0.21$). The nature of the two-way interaction is presented in Fig. 4.

Table 4
The effect of failure severity and brand reputation on satisfaction and behavioral intentions.

Sources	MANOVA			Univariate		
	Wilks' lambda	F value	Sig.	df	Satisfaction	Behavioral intentions
Main effects						
Failure severity (FS)	0.87	19.25	0.00	1	7.04 (0.01)	37.62 (0.00)
Brand reputation (BR)	0.87	15.69	0.00	1	3.46 (0.06)	31.4 (0.00)
Interaction effect (FS X BR)	0.97	3.46	0.03	1	5.84 (0.02)	2.80 (0.09)

Customer satisfaction and behavioral intention means by failure severity and by brand reputation condition

Dependant variable: customer satisfaction			
		Brand reputation	
		Less	High
Failure severity	Low	2.31 (1.40)	3.18 (1.80)
	High	2.26 (1.65)	2.15 (1.62)

Dependant variable: behavioral intentions			
		Brand reputation	
		Less	High
Failure severity	Low	2.55 (1.21)	3.71 (1.34)
	High	1.85 (1.06)	2.47 (1.41)

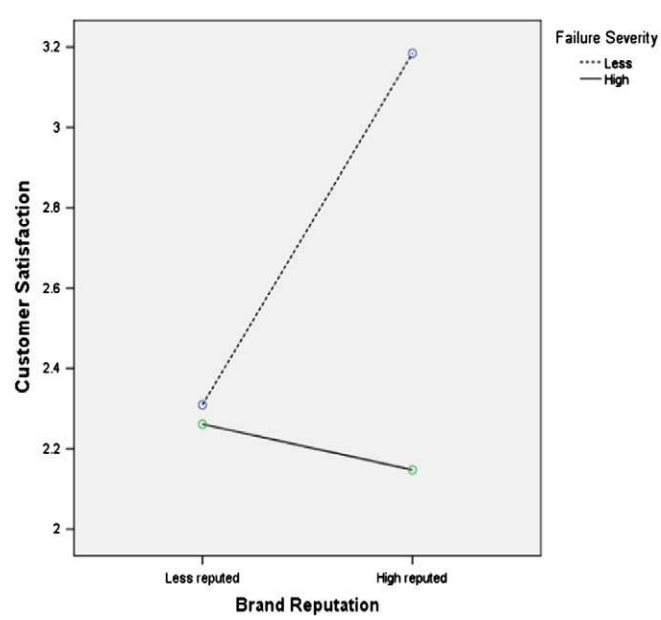

Fig. 2. Interaction of failure severity and brand reputation on customer satisfaction.

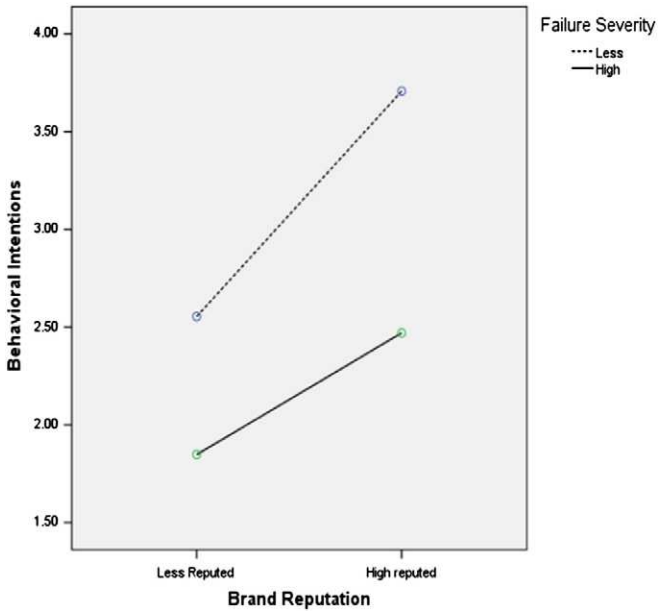

Fig. 3. Interaction of failure severity and brand reputation on behavioral intentions.

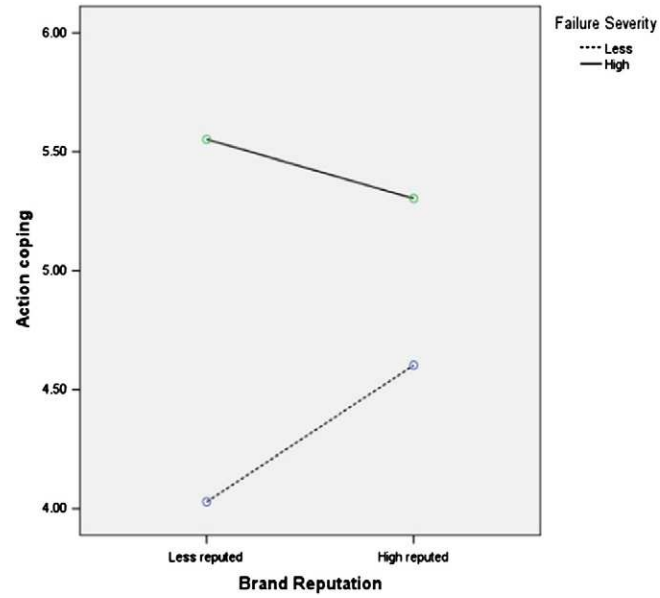

Fig. 4. Interaction of failure severity and brand reputation on action coping.

The results show that for less severe condition, respondents reported greater rational thinking for high reputed brand (M = 4.46) than less reputed brand (M = 4.09) (simple slope = 0.18, $p < 0.05$). However, no significant difference was observed in the cell means of rational thinking in high severe condition for high reputed (M = 3.79) and less reputed brands (M = 4.07) (simple slope = −0.10, $p = 0.22$). Fig. 5 presents the form of interaction on rational thinking.

Examination of cell means showed that respondents in less severe condition reported greater positive thinking for high reputation brand (M = 4.20) than less reputed brand (M = 3.74) (simple slope = 0.16, $p < 0.05$). The relationship between high severe-high reputation (M = 2.98) and high severe-less reputation groups (M = 3.44) was moderately significant (simple slope = −0.14, $p < 0.10$). We plotted the two-way interaction for positive thinking in Fig. 6.

Respondents reported greater emotional venting for high reputed brand (M = 3.99) than less reputed brand (M = 3.25) (simple slope = 0.29, $p < 0.01$) only in less severe condition. No significant difference was observed in high severe failure condition (simple slope = −0.01, $p = 0.92$). The interaction is plotted in Fig. 7. These results provide partial support for H4.

4.3.3. Testing H3

The mediation effect of coping strategies was tested using ANOVA and ANCOVA models as proposed by Baron and Kenny (1986). Table 6 presents the results of the mediation analysis. The findings revealed that rational thinking, positive thinking, avoidance and denial mediated the relationship of failure severity and brand reputation with customer satisfaction.

5. Discussion

This study focuses on understanding the influence of failure severity and brand reputation on customer coping strategies, satisfaction, and behavioral intentions in the context of airline services. Overall, the results of the study provide empirical support for the process-oriented framework that links coping to service failure, brand reputation, and customer outcomes of satisfaction and behavioral intentions. Building on the stress-and-coping theory (Lazarus & Folkman, 1984), the present study provides a new perspective into service failures by emphasizing the importance of coping process as a psychological response to stressful situations. We demonstrate that customers use various coping strategies such as action coping, positive thinking, and support seeking to manage negative experiences and stress associated with service failures. Moreover, for high severe service failures while customers employ problem solving coping strategies such as action coping and instrumental support seeking, they rely on emotion-based coping strategies such as rational thinking, positive thinking, avoidance and denial for less severe service failures. Moreover, these emotion-focused coping strategies were observed to mediate the relationship between service failure and customer satisfaction. Brand reputation was found to moderate the relationship between service failure and customer satisfaction such that the buffering effect was observed for less severe failures. In spite of the lower satisfaction levels, the findings reveal that the respondents reported higher levels of behavioral intentions for high reputed brands than less reputed brands when they encounter high severe failure.

Table 5
Effect of failure severity and brand reputation on the coping strategies.

Sources	MANOVA			Univariate								
	Wilks' lambda	F	Sig.	df	Action coping	Rational thinking	Positive thinking	Emotional venting	Emotional-support	Instrumental-support	Avoidance	Denial
Main effects												
Failure severity FS	0.74	10.8	0.00	1	54.76 (0.00)	5.44 (0.02)	15.18 (0.00)	1.90 (0.17)	2.05 (0.15)	4.67 (0.03)	16.33 (0.00)	5.46 (0.02)
Brand reputation BR	0.96	1.4	0.20	1	1.17 (0.28)	0.08 (0.78)	15.10 (0.00)	5.10 (0.025)	0.00 (0.98)	0.60 (0.44)	0.03 (0.86)	0.86 (0.35)
Interaction effect												
FS X BR	0.92	2.85	0.01	1	7.44 (0.00)	4.72 (0.03)	5.78 (0.02)	5.76 (0.02)	1.13 (0.29)	0.02 (0.89)	0.56 (0.45)	2.58 (0.11)

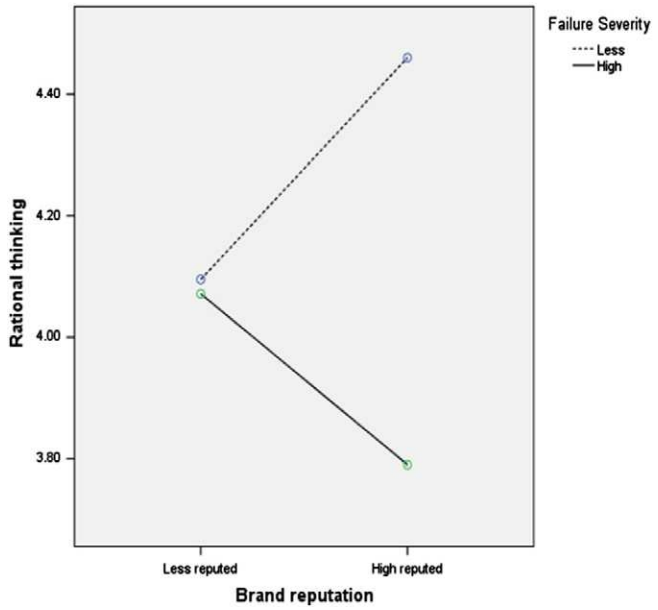

Fig. 5. Interaction of failure severity and brand reputation on rational thinking.

6. Implications

This study has important implications for the service managers. The present research demonstrated that customers use a repertoire of coping strategies in managing service failures. It was found that customers perceive vulnerability in high severe failure and engage in action coping and instrumental coping strategies to resolve the problem. Thus, service managers should devise recovery strategies that encourage customers to discuss the problem with the organization and find acceptable solutions. This would bring the failure to the service providers' attention helping them to adapt, monitor and maintain the service delivery process. Another strategy in such failures would be to involve the customers in the recovery process. Co-participation not only attunes the negative failure outcomes, but also provides customer with a successful recovery experience. It engages the customer in the recovery process and contributes to customer relationship building and maintenance (Dong, Evans, & Zou, 2008).

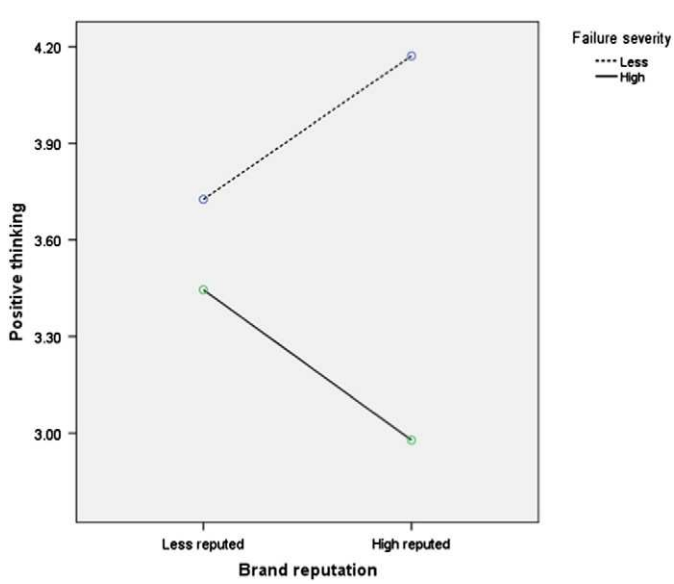

Fig. 6. Interaction of failure severity and brand reputation on positive thinking.

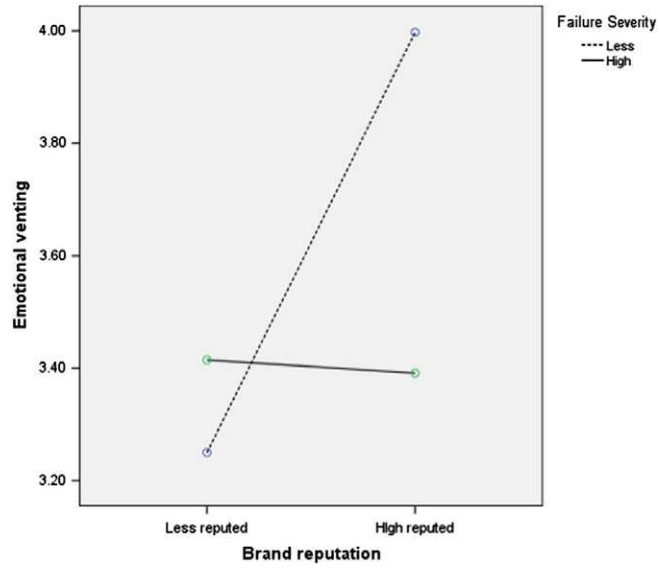

Fig. 7. Interaction of failure severity and brand reputation on emotional venting.

On the contrary, customers experiencing less severe failures were found to engage in rational-thinking, positive thinking, avoidance and denial coping strategies. Although these coping strategies lead to favorable customer outcomes, behavioral research has shown that suppressing feelings lead to depression and anxiety (Amstadter, 2008). In such cases, service managers should devise recovery strategies that provide emotional support and show concern and empathy that customers are seeking. Understanding the role of brand reputation is important given its influence on evaluation of service failure and recovery. Our findings show that the positive quality associated with high reputation brands influences the service failure evaluations resulting in favorable post-failure satisfaction and behavioral intentions. Thus, brand-building can be used as a recovery strategy to influence repurchase intentions. Though the results suggest that high brand reputation fails to shield the detrimental effects of high severe failures, it does not prevent customers from re-patronizing the brand.

Our study findings indicate that customers reported higher levels of behavioral intentions for high reputation brand than less reputation brand for both less severe and high severe failures. This finding points to the importance of building brand reputation as a short-term benefit to diminish failure outcomes. To mitigate high severe failures, explanation can be used to reduce helplessness and prevent customers from engaging in negative word-of-mouth (Gelbrich, 2010). For this, service personnel should take a proactive approach to identify and fix the problem. In particular, employees should try to foster rapport with the customers and reduce anger and frustration with the failure.

7. Limitations and further research

This research has several limitations. First, while we focused on the eight coping strategies, service failures may display other coping strategies such as self-blame, humor, acceptance and emotional processing (Zuckerman & Gagne, 2003). Future research should incorporate more coping strategies in the framework to examine the service failure outcomes. Second, in this research we assessed service failure severity and brand reputation in examining the customer outcomes. Future research should investigate the role of complaint behavior and perceived recovery efforts to enhance the external validity of the findings. Third, the study was carried out in Airline context which required some customer-employee interaction. As such generalization of this research should be handled with caution. Future research should broaden the understanding of failure severity and brand reputation by

Table 6
Mediation analysis of coping strategies.

	ANOVA satisfaction	ANOVA RT	ANCOVA	ANOVA PT	ANCOVA
Failure severity: FS	6.92 (0.00)	5.47 (0.02)	5.43 (0.02)	15.54 (0.00)	2.80 (0.09)
Brand reputation: BR	3.60 (0.05)	0.083 (0.77)	3.51 (0.06)	0.01 (0.92)	3.90 (0.05)
FS X BR	5.73 (0.02)	4.8 (0.03)	4.5 (0.03)	5.61 (0.02)	3.30 (0.07)
Covariate: RT			3.98 (0.05)		
Covariate: PT					17.19 (0.00)
	ANOVA satisfaction	ANOVA avoidance	ANCOVA	ANOVA denial	ANCOVA
Failure severity: FS	6.92 (0.00)	16.19 (0.00)	3.67 (0.05)	5.46 (0.02)	1.74 (0.18)
Brand reputation: BR	3.60 (0.05)	0.02 (0.88)	3.90 (0.05)	0.86 (0.35)	3.51 (0.06)
FS X BR	5.73 (0.02)	0.518 (0.47)	6.50 (0.01)	2.58 (0.11)	3.21 (0.07)
Covariate: avoidance			7.35 (0.00)		
Covariate: denial					366.45 (0.00)

examining their effects in other service categories such as self-service technologies.

Fourth, future studies should incorporate negative emotions in the framework as prior studies indicate that negative emotions are linked to different coping strategies (Yi & Baumgartner, 2004). Fifth, we assessed customer satisfaction with a single item as overall satisfaction. Future research should use multiple item measure as it increases the validity of the measure. Finally, this study examined the role of brand reputation in evaluation of service failure outcomes. However, we did not consider other constructs such as past failure experience, attribution of the failure and relationship quality that could possibly influence the relationship. Future research should consider these constructs to identify the boundaries of the effect of brand reputation on failure outcomes.

Appendix 1

High reputed brand condition (less reputed brand condition).

Imagine that you have booked tickets in the AAA airline to meet your friend for a relaxing trip. The airline was recently awarded 5-Star Ranking by SKYPAX (3-star Ranking). The SKYPAX recognizes the airline for its product and service delivery quality achievement. This airline company is well-known for on-time arrival and departure (some delays in arrival and departure). Further, this airline provides excellent in-flight services (average in-flight services).

High severe failure condition (less severe failure condition).

You arrive at the airport and wait in the check-in line. Soon it is your turn, and you tell the staff your flight destination. However, the staff responded that this flight is canceled (delayed by 2 hours). On further enquiring, the staff member was unapologetic (was apologetic) and rude (polite).

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